

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
MASTER OF COMPUTER APPLICATIONS Semester : 1
23MX13 DATA STRUCTURES

Time : 1 Hour 30 mins

Maximum Marks : 50

- 1) i) Differentiate linear and non-linear data structures. Give an example. (2)
- ii) If $f(n) = O(g(n))$ and $h(n) = O(g(n))$. Is $f(n) = g(n)$? Justify your answer. (2)
- iii) If $f(n) = O(g_1(n))$ and $h(n) = O(g_2(n))$. What is $O(f(n).h(n))$? Justify your answer. (2)
- iv) Write an efficient algorithm to find x^n using repeated multiplications. What is the complexity? (4)
- 2) Find the running time of the following program segments. (10)

i) for(int i = n; i > 0; i /= 2) for(int j = 1; j < n; j *= 2) for(int k = 0; k < n; k += 2) printf("%d %d %d", i, j, k);	ii) for(int i = 0; i < n; i ++) { for(int j = 0; j < n; j *= 2) printf("**"); break; }
iii) for(i=n/2; i<=n; i=i+1) for (j= 1; j+n/2 <=n; j=j+1) for (k=1; k<=n; k=k*2) count ++;	iv) for(i=0; i<=n; i=i+1) for (j= i; j <= i*i; j=j+1) if(j%2 ==0) { for (k=0; k< j; k=k +1) count ++; }
v) int fun(int n) { int count = 0; for (int i = n; i > 0; i /= 2) for (int j = 0; j < i; j++) count += 1; return count; }	

- 3) i) What is abstraction? What is an ADT? Define String ADT. (4)
- ii) Write the recurrence relation describing the following code. Assume the complexity of the function merge(int, int, int) is $O(n)$. Solve the recurrence relation and indicate the complexity of the code in Big Oh notation. (6)

```
algorithm mergesort(low, high) {
    if (low < high) {
        mid = (low + high) / 2
        mergesort(low, mid)
        mergesort(mid+1, high)
        merge(low, mid, high) } }
```

- 4) i) Solve $T(n)=4T(n/2)+n$; $T(1)=1$. Verify the solution using Master's theorem. (5)
- ii) Write an i) iterative ii) recursive algorithm to find the factorial of a given number.
What is the complexity? (5)
- 5) i) What are the ways of representing three-dimensional arrays in memory? Give suitable examples. Write the corresponding addressing functions. (5)
- ii) A Toeplitz (or diagonal-constant) matrix is a matrix in which each descending diagonal from left to right is constant, i.e., all elements in a diagonal are same. Suggest an efficient storage representation for storing a toeplitz matrix and a symmetric matrix. Write the corresponding addressing function. (5)

10	4	3	2	1
4	10	4	3	2
3	4	10	4	3
2	3	4	10	4
1	2	3	4	10

Toeplitz Matrix

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
 MASTER OF COMPUTER APPLICATIONS Semester : 1
 23MX13 DATA STRUCTURES
 Test-2

Time : 1 Hour 30 mins

Maximum Marks : 50

Date: 21-12-2023

INSTRUCTIONS:1. Answer **ALL** questions. Each question carries 25 Marks.

2. Course Outcome Table:

Qn.1

CO3

Qn.2

CO4

- 1) a. i) Construct a binary tree whose inorder traversal sequence is G D H B E I A F J C (2)
 and the preorder traversal sequence is A B D G H E I C F J
- ii) Construct a strictly binary tree where the non-leaf nodes have 2 children and the (3)
 preorder traversal sequence as 30, 20, 15, 5, 18, 25, 40, 35, 50, 45, 60 and
 postorder traversal sequence is 5, 18, 15, 25, 20, 35, 45, 60, 50, 40, 30
- b. i) Identify suitable data structures for the following: (2)
1. Priority Queue
 2. Representing contacts in social media
 3. Representing mathematical expressions
 4. Representing the file directories.
- ii) Construct a binary search tree of level 3 with the following sequence of numbers: (3)
 10, 12, 5, 4, 20, 8, 7, 15 and 13.
- iii) What are the different ways of representing a Graph in memory? illustrate with (3)
 the graph shown in Fig. 1.

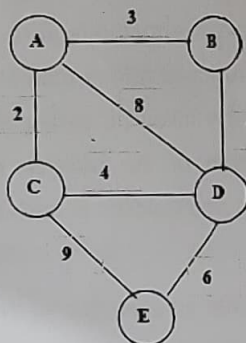


Fig.1

- c.i) Write an algorithm for traversing a Graph using Depth First Search method. Trace (6)
 the algorithm with the graph given in Fig. 2.

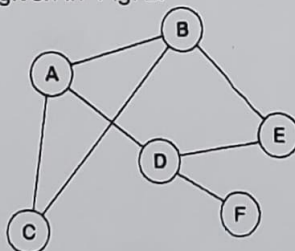


Fig.2

- c.ii) Write an algorithm to perform deletion in Binary Search Tree. Consider the tree (6) given in fig. 3, show the tree after deleting 8, 12, 6 and 10.

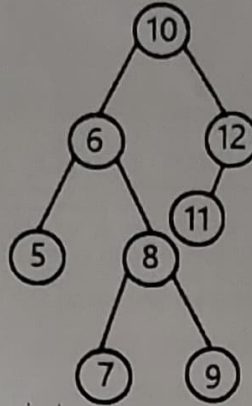


Fig.3

- 2) a. i) Illustrate Heap sort with the following elements: 34, 67, 89, 12, 45, 65, 90. What is the complexity? (5)
- b.i) Formulate linear search algorithm. Trace the algorithm on a sample input. Indicate the best and worst case complexity. (4)
- b. ii) Write an algorithm for Insertion Sort. Applying the algorithm to sort the following list of elements 12, 25, 3, 45, 56, 34, 70, 27, 189, 59, 75. Show the results of the intermediate passes, Analyze the best and worst case complexity. (4)
- c. What is Hashing? What is the advantage and limitation of using Hashing to search an element? Explain the methods of resolving collisions. Consider a hash table of size 11 with hash function $h(k) = k \bmod 11$. Draw the table that results after inserting in the given order, the following values: 9, 26, 50, 15, 2, 21, 36, 22 and 31. Resolve collisions using each of the collision resolving techniques discussed. Describe insertion, searching and deletion in each of the methods discussed (12)
- (OR)
- c. For a binary tree, write an algorithm for the following: (12)
- To count number of leaf nodes.
 - To find the level of a node with the given value
 - To check whether two trees are structurally identical.
- Example for each.*

./END/

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
 MASTER OF COMPUTER APPLICATIONS Semester : 1
 23MX13 DATA STRUCTURES

Time : 1 Hour 30 mins

Maximum Marks : 50

INSTRUCTIONS:1. Answer **ALL** questions. Each question carries 25 Marks.

2. Course Outcome Table:

Qn.1	CO1	Qn.2	CO2
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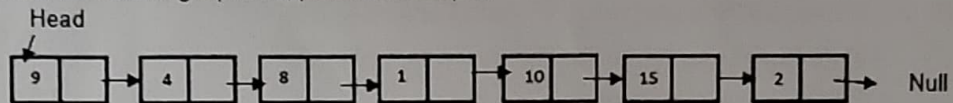
- 1) a. i) What are the characteristic properties of an algorithm? (2)
 ii) Write an algorithm to find the quotient of m/n given m and n . Do not use division operator. (3)
 What is the complexity of your algorithm?
- b. i) Differentiate linear and non-linear data structures. Give examples. (3)
 ii) Define Big O. How is it related to algorithm computational complexity? (5)
 If $f(n) = O(g_1(n))$ and $h(n) = O(g_2(n))$, answer the following questions. Justify your answer
- a) What is $O(f(n).h(n))$?
 b) What is $O(f(n)+h(n))$?
 c) If $g_1(n) = g_2(n)$, Is $f(n)=h(n)$?
- c. i) Indicate the complexity of the following code segments using Big O notation. (6)
- 1) for (j = n/2; j ≤ n; j++)
 for (k = 1; k ≤ n; k = 2 * k)
 for (m = 1; m ≤ n ; m = m * 2)
 printf("#")

2) for(i=0; i<=n; i=i++)
 for (j= i; j <= i*i; j=j++)
 if(j%2 ==0) {
 for (k=0; k<j; k=k++)
 count++;}
 }
- 3) int fun(int n){
 int count = 0;
 for (int i = n; i > 0; i /= 2)
 for (int j = 0; j < i; j++)
 count += 1;
 return count; }
- 4) for(int i = 0; i < n; i ++) {
 for(int j = 0; j < n; j *= 2)
 printf("***");
 break;
 }
- ii) Solve the following recurrence relation, using substitution method (6)
 $T(n) = 2T(n/2) + n$;
 $T(1) = 1$.
 Using Master Theorem, verify the solution.
- 2) a. Describe static and dynamic memory allocation. What are their advantages and limitations? (5)
 b. i) Write a code snippet to count the number of nodes in a singly linked list. (5)

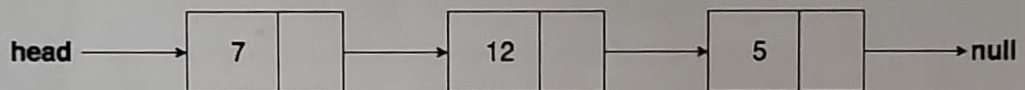
ii) What is the functionality of the following code snippet? (2)

```
count = 0;
while (head != NULL)
{
    if (count % 2 == 0)
        print(data);
    count++;
    head = head->next;
}
```

Given the following input list, predict the output.

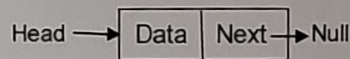


iii) Consider singly linked list in the format shown below: (3)



Fill up the blank lines in the following code segment to print the sum of elements in the given linked list.

```
struct Node {
    int data;
    struct Node* next;
};
int sum = 0;
while (_____) {
    _____
    print (sum);
}
```



c. i) How are two dimensional arrays represented in memory? Write the corresponding addressing functions. Extend this to represent the k-dimensional array. (4)

ii) Suggest an efficient way of storing the tri-band matrix and write the corresponding addressing function. A sample band matrix of order 4 x 4 is shown below: (4)

10	5	0	0
9	6	5	0
0	1	7	3
0	0	19	2

iii) A Toeplitz matrix is a diagonal constant matrix i.e. all elements along a diagonal have the same value. Following is an example of Symmetric Toeplitz matrix. Suggest an efficient representation for storing Symmetric Toeplitz matrix. Write the corresponding addressing function. (4)

1	2	3	4	5
2	1	2	3	4
3	2	1	2	3
4	3	2	1	2
5	4	3	2	1

./END/

No of Pages : 3

Course Code : 23MX13

Roll No:

(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, JUNE 2024

MCA Semester: 1

23MX13 DATA STRUCTURES

Time : 3 Hours

Maximum Marks : 100

INSTRUCTIONS:

1. Answer ALL questions. Each question carries 25 Marks.
2. If any information is missing, make appropriate assumptions and mention the same.

1. a) Define asymptotic notations for lower and upper bound complexity of algorithms.

What is the complexity of the following code?

(5)

```
sum = 0
```

```
for ( i = 0; i < n/2; i ++)
```

```
    for ( j = 1; j < 8 + i ; j ++)
```

```
        for ( k = 0; k < 8 ; k ++)
```

```
            print("$")
```

- b) i) What are the properties of an algorithm? Given an array of size N-1 with distinct integers in the range of 1 to N, write an algorithm to find the missing element. What is the time and space complexity of your algorithm? (4)

- ii) For a task, different algorithms are written and their complexity are $O(n)$, $O(\log n)$, $O(n \log n)$, $O(n^2)$, $O(2^n)$. Which is good and which is worst? Justify your answer. (4)

- c) i) Solve the recurrence relation $T(n) = T(n/2) + n$ (6)

- ii) Write the recurrence relation corresponding to the following algorithm. Solve the same to find the complexity. (6)

Algorithm TOH (n, x, y, z)

```
if (n >= 1) {
```

```
    TOH ( (n-1), x, z, y)
```

```
    x = y
```

```
    TOH ((n-1), z, y, x)
```

```
}
```

2. a) Write an algorithm to find the sum of the elements stored in a linked list. Insert a node at the end of the list and store the sum. (5)
- b) i) A 2- dimensional array $A(5 \times 4)$ of integers is stored starting from location 1000. An integer occupies 2 bytes. Calculate the address of the element $A[3, 2]$ if the array is stored using i. row major order and ii. Column major order. Write the addressing functions. (4)
- ii) What are sparse matrices? Suggest an efficient storage representation of sparse matrix. Illustrate with an example. (4)
- c) i) Consider a mathematical expression with $()$, $[]$ and $\{ \}$. Parentheses may be nested. Write an algorithm to check for the well formedness of parentheses in the given expression. (6)
- ii) Write algorithms for the primitive operations on queue. (6)
3. a) How are binary trees represented in storage? Illustrate with suitable example. Which type of representation is efficient for i. complete binary tree and ii. Skew tree? Justify your answer. (5)
- b) i) What are the different ways of traversing a binary tree? Indicate the order in which the nodes are visited. Write the traversal sequences for the tree shown in Fig 1. (6)

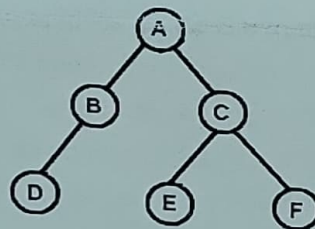


Fig 1

- ii) How are graphs represented in storage? Show the representations for the graph shown in Fig 2. Indicate the advantage and limitation of each representation. (6)

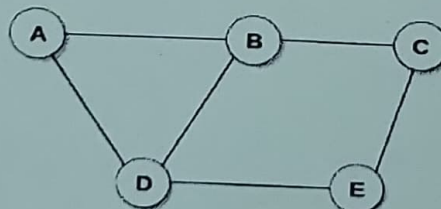


Fig 2

No of Pages : 3

Course Code : 23MX13

- c) i) What is a Binary Search Tree? Construct a Binary Search Tree by inserting the elements 56, 34, 99, 12, 39, 78, 45. Delete 34 and show the resultant tree. Write algorithms for insertion and deletion operation.

(OR)

- ii) Write algorithms for Breadth first and depth first traversal on graphs. Trace the algorithms on the graph shown in Fig 2.

4. a) Sort the following list of elements using Radix Sort : 56, 34, 99, 12, 39, 78, 45, 5, 122. What is the complexity of Radix Sort. (5)
- b) Write algorithm for linear and binary Search. Indicate the best and worst case complexity. Illustrate with suitable examples. (8)
- c) i) 1) Write algorithm for Insertion Sort. What is its complexity? Sort 56, 34, 99, 12, 39, 78, 45 and show the results of intermediate passes also. (6)
- 2) Sort the list 56, 34, 99, 12, 39, 78, 45, 5, 122 using heap sort. What is the complexity of heap sort? (6)

(OR)

- ii) Consider hash function for n is $n \% 20$ and second hash function, where applicable, is

$(n \% 7) + 1$. Suppose separate chaining is used for collision resolution. Beginning with an empty hash table, insert the following elements 8, 38, 3, 5, 28, 18, 65, 83 into the hash table. How many of the inserted keys will be looked at when 48 is searched?

- 1) Suppose linear probing is used for collision resolution. What is the answer to the above questions?
- 2) Suppose double hashing is used for collision resolution. What is the answer to the above questions?

/END/

FD/RSP

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
MASTER OF COMPUTER APPLICATIONS SEMESTER : 1
23MX13 DATA STRUCTURES
TUTORIAL-2

MARKS: 25 Marks

- 1) Write an algorithm to perform the following:
 - k) Count the number of nodes in a singly linked list
 - l) Concatenate the shorter list at the end of a longer list
 - m) Reverse the singly linked list
 - n) Delete the node at the end of the list
 - o) Delete the node at the specified position.
- 2) Write algorithms to i) Push and ii) Pop an element from a stack using an array
- 3) Write an algorithm to reverse a string using the stack.

Maximum Marks : 100

No of Pages : 3

Course Code : 23MX13

Roll No: 23mx304

(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester : 1

23MX13 DATA STRUCTURES

Time : 3 Hours

Maximum Marks : 100

INSTRUCTIONS:

1. Answer ALL questions. Each question carries 25 Marks.
2. Subdivision (a) carries 5 marks each, subdivision (b) carries 8 marks each and subdivision (c) carries 12 marks each.
3. If any data is found missing, suitable assumptions can be made with proper justification.
4. Course Outcome :

Table

Qn.1 CO1

Qn.2 CO2

Qn.3 CO3

Qn.4 CO4

1. a) i) What is an ADT? What is its need? Define Queue ADT. (5)
- b) What are the characteristics properties of an algorithm? Write an algorithm for printing the first n elements in the Fibonacci sequence. Show that your algorithm possess all the properties of an algorithm. (8)
- c) i) Indicate the complexity of the following code segments using Big O notation. (4)

1) for (j = 1; j ≤ n; j++)
 for (k = n; k > 1; k = k/2)
 for (m = 1; m ≤ n; m = m * 2)
 printf("\$")

2) for(j = 1; j < n; j++)
 for(k = 1; k < n; k++) {
 print("\$")
 break;
 }
- ii) Define Big O notation. Write the significance of Big O notation. Solve the following recurrence relation, using substitution method and write the complexity using Big O notation (8)

$$T(n) = T(n-1) + T(n-2) + c$$

$$T(1) = 1$$
2. a) What are the various ways of representing two dimensional matrix in storage? Explain with suitable examples. Write the corresponding addressing functions. Suggest an efficient representation for storing an upper triangular matrix and write the addressing function. (5)
- b) i) Write an algorithm using Stack ADT calls to recognize strings with balanced parentheses and square brackets. Trace the algorithm with strings ([()], (()]) (4)
- ii) Implement a queue using a singly linked list. (4)

- c) i) Consider the ADT Set that represents a collection of Integers. The ADT should support standard set operations such as the following:

- i. $S = \text{init}()$ /* Initialize S to the empty Set */
- ii. $\text{isSingleton}(S)$ /* Return True if and only if S is an empty Set */
- iii. $\text{isMember}(S, a)$ /* Return True if and only if a is a member of the Set S */
- iv. $S = \text{addElement}(S, a)$ /* Add a to the Set S. If a is already in S, there will no change in S */
- v. $S = \text{delElement}(S, a)$ /* Remove a from the Set S. If a is not a member of S, there will no change */
- vi. $\text{Print}(S)$ /* Print the elements of S */

Implement the Set ADT using Linked Lists

(12)

(OR)

- ii) Formulate algorithms for the following

- 1) Given two sorted list, merge them as a singly linked list
 - 2) Given a doubly linked list and a key, print the data stored in the predecessor and successor nodes of the key and delete the key element. If the key element is not present, print an appropriate message.
3. a) Write the sequence in which nodes are traversed in inorder, preorder and postorder traversal on Binary Trees. Draw a binary tree whose inorder traversal sequence is H D J B E A F J C G and post order traversal sequence is H J D E B J F G C A. (5)

Draw a Binary Tree in which all the non-leaf nodes have two children and whose pre order traversal sequence is 100, 34, 16, 9, 8, 38, 11, 4, 81 and post order traversal sequence is 34, 9, 11, 4, 38, 81, 8, 16, 100.

- b) i) Define heap data structure. Why is heap a preferred data structure to implement priority queue? Illustrate the max heap construction with the keys 34, 56, 17, 12, 8, 3, 2, 1, 0. What is the time complexity for inserting an element into the heap? (5)
 - ii) Draw a) directed and b) undirected graph with 5 vertices and 6 edges. Show their representation using Adjacency matrix and Adjacency list (3)
- c) i) Write algorithms for Breadth First Search and Depth First Search on Graphs. Tracing the algorithms, write the traversal sequence for the graph given in fig.1. Modify any one of the algorithms to check whether the given graph is connected. (12)

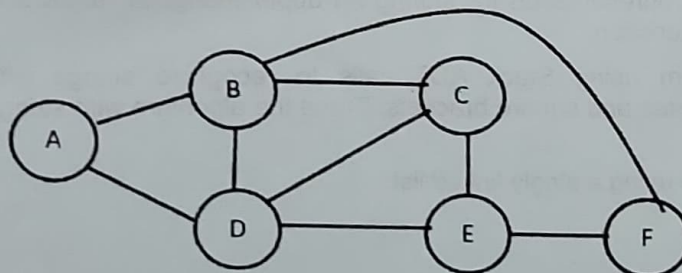


Fig.1

(OR)

- ii) What is a Binary Search Tree? Formulate algorithms for insertion and deletion on a Binary Search Tree. Create a Binary Search tree of level 4 with the following elements : 45, 65, 78, 90, 23, 30, 10, 50, 5, 10, 60, 12
4. a) Illustrate heap sort and radix sort with the following numbers. 45, 65, 78, 90, 23, 30, 10, 50. Show the result of the intermediate passes. What is the complexity of these algorithms? (5)
- b) i) Formulate an algorithm for binary search. When Binary Search algorithm is applied to search for keys 67 and 15 in the list of elements 2, 4, 12, 22, 37, 39, 45, 48, 52, 64, 67, 69, 80, in which order elements will be compared with the keys. What is the time complexity for successful and unsuccessful search? (4)
- ii) Formulate algorithm for Bubble sort. What is the complexity of Bubble sort algorithm. (4)
- c) What is Hashing? Describe any two methods for finding the hash address. What is meant by Collision in hashing? Describe any two ways of resolving collisions. The following elements are to be entered in the given order in a hash table of size 11: 12, 31, 63, 89, 90, 78, 55 and 21. Apply a suitable hash function. Show the contents of the table. Illustrate collision resolution methods described. For each of the methods, how many comparisons are to be made in locating the elements: 31, 89 and 56. (12)

FD/RL

/END/